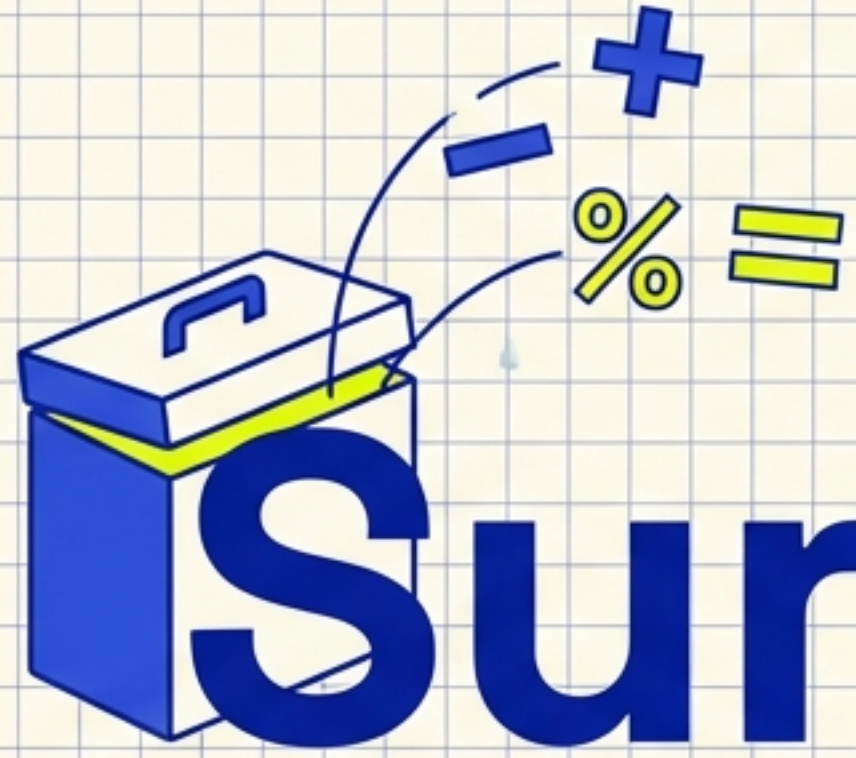




Math LZK Survival Guide

Equations & Percentage Calculations

THE MISSION

MISSION CHECKLIST



The Goal: Get you 100% ready for the 02-03-2026 LZK.



The Promise: Simple explanations. No jargon without translation.



The Toolkit: Equations, Percentages, and Real-World Logic.



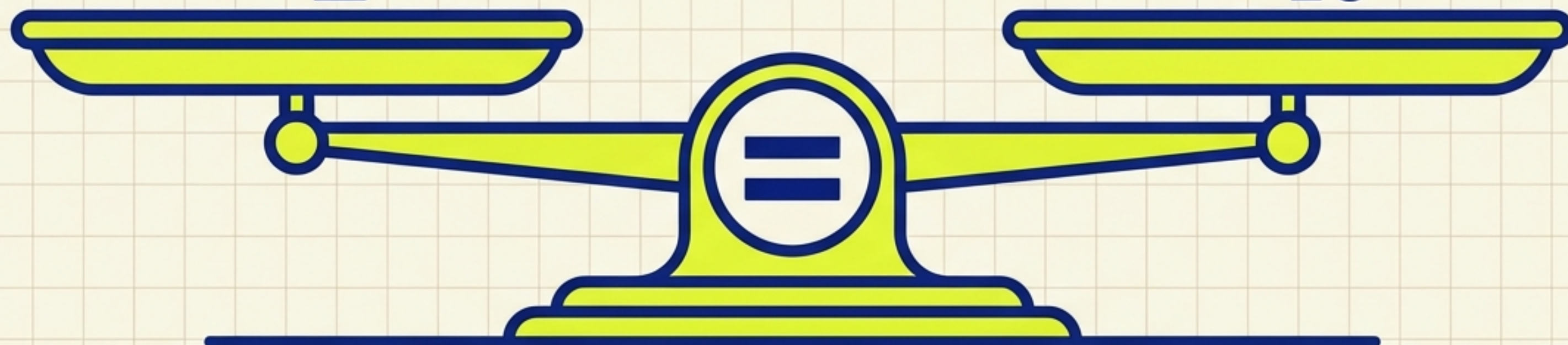
Equations: The Golden Rule

JetBrains Mono

T_L

JetBrains Mono

T_R



The Rule: Whatever you do to one side, you **MUST** do to the other.

Objective: Isolate the variable (Get x alone).

THE MOVES: INVERSE OPERATIONS

To remove +3

 Subtract 3 (-3)

To remove -3

 Add 3 (+3)

To remove $\cdot 3$

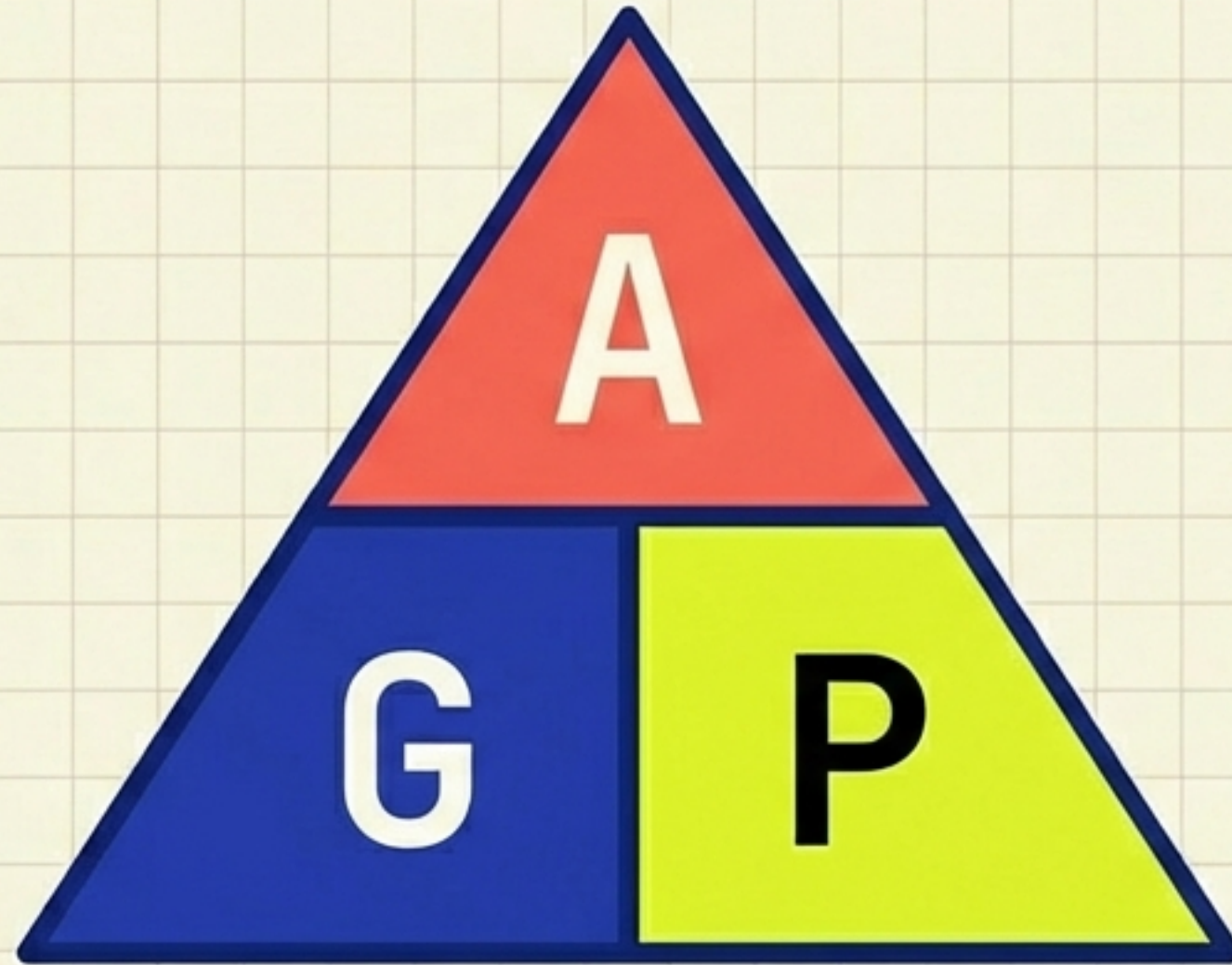
 Divide by 3 ($:3$)

To remove $:3$

 Multiply by 3 ($\cdot 3$)

$$\begin{array}{r}
 3 \cdot x + 5 = 11 \\
 | -5 \\
 3x = 6 \\
 | :3 \\
 x = 2
 \end{array}$$

Percentages: The Magic Triangle



G (Grundwert) = The Whole / Base

A (Anteil) = The Part / Slice

P (Prozentsatz) = The Rate
(decimal! 20% = 0.2)

Cover the letter you need to find:

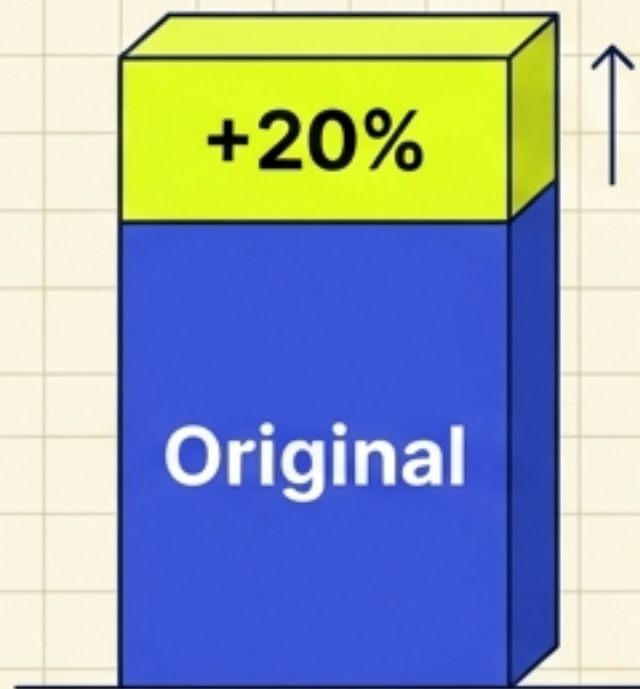
Cover **A** → **G** · **P**

Cover **G** → **A** / **P**

Cover **P** → **A** / **G**

The Language Trap: 'Um' vs. 'Auf'

UM / BY

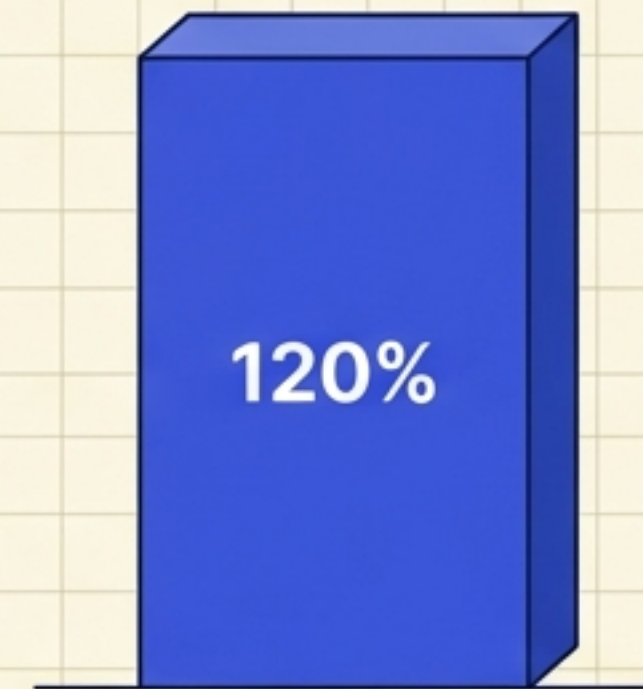


Focus: The Change

Example: Increases BY 20%

Math: Add to original ($100\% + 20\%$)

AUF / TO



Focus: The Result

Example: Increases TO 120%

Math: This IS the new value (1.20)

VSN Pro Tip: Convert 'Um' problems into 'Auf' multipliers for speed!

►► Speed Math: The Change Factor



Increase by $p\%$

Multiply by $(1 + p/100)$

USB Stick (€12) +4%
→ $12 \cdot 1.04 = 12.48$



Decrease by $p\%$

Multiply by $(1 - p/100)$

Pants (€64) -25%
→ $64 \cdot 0.75 = 48$

Don't Get Tricked: Points vs. Percent

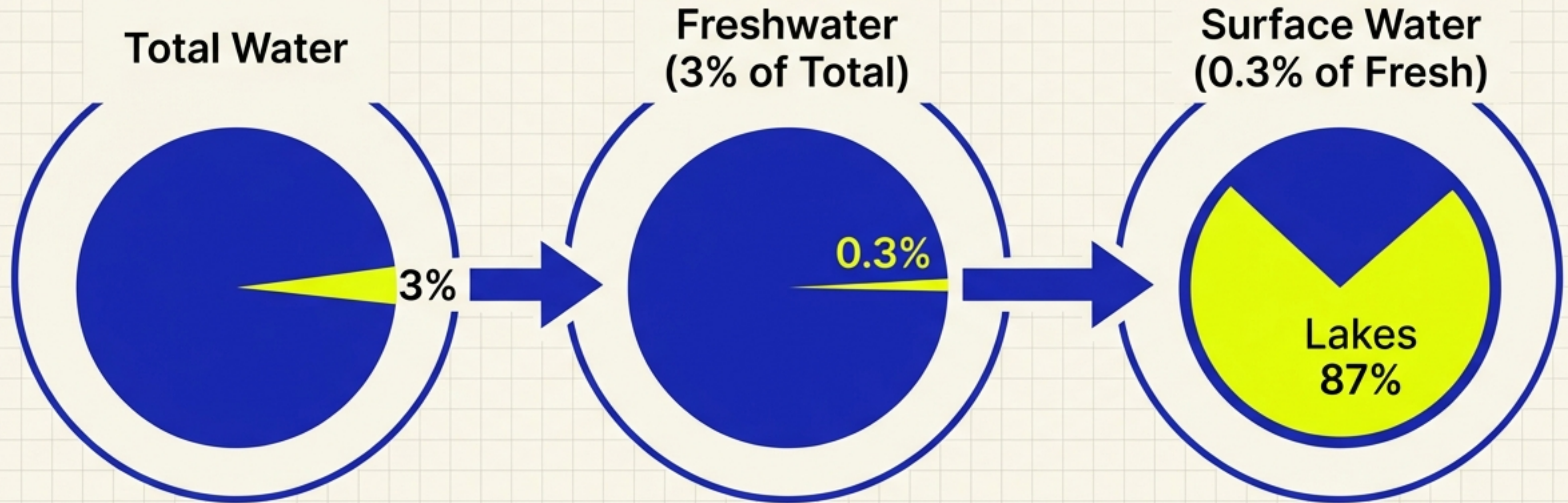


Scenario: Interest rate goes from 5% to 6%.

~~It went up by 1%.~~

Percentage Points (Absolute)	Percent Change (Relative)
$\$6 - 5 = 1$	$(6 - 5) / 5 = 0.20$
1 Percentage Point	20% Increase

→ Real World: Nested Percentages (Water)



Math: $1 \cdot 0.03 \cdot 0.003 \cdot 0.87 = 0.000783\%$

Multiply the factors to tunnel down.

Real World: Election Night Statistics

Party	2008 Result	2013 Result
SPÖ	39.39%	23.81%

Points Lost

$$39.39 - 23.81 = 15.58$$

Percentage Points

Relative Loss

$$(23.81 - 39.39) / 39.39$$

$\approx -39.5\%$ Relative Loss

Note the massive difference between losing 15 points and losing 40% of voters.

Real World: Money Matters

Shop Receipt

Gross Price
- Rabatt (Sale)

= New Price
- Skonto (Cash Discount)

= Final Payment

Vocab Decoder

Rabatt = Sale

Brutto = With Tax

Skonto = Pay-fast bonus

Netto = Without Tax

The VAT Trick (20%)

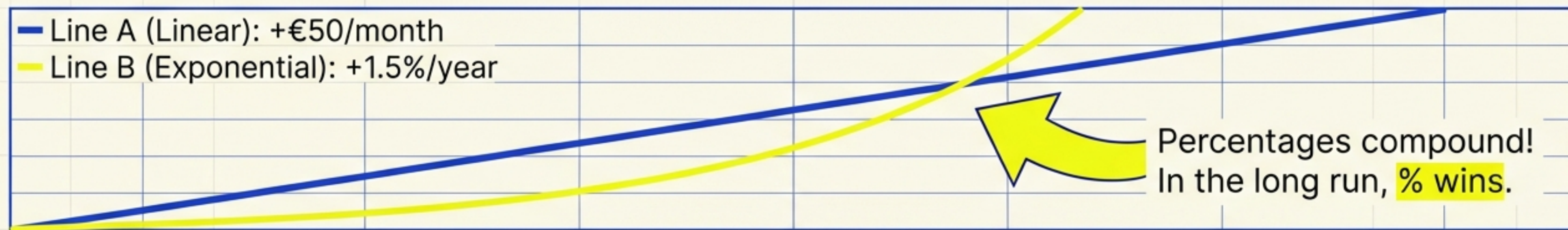
Netto $\cdot 1.20 \rightarrow$ Brutto

Brutto $/ 1.20 \rightarrow$ Netto

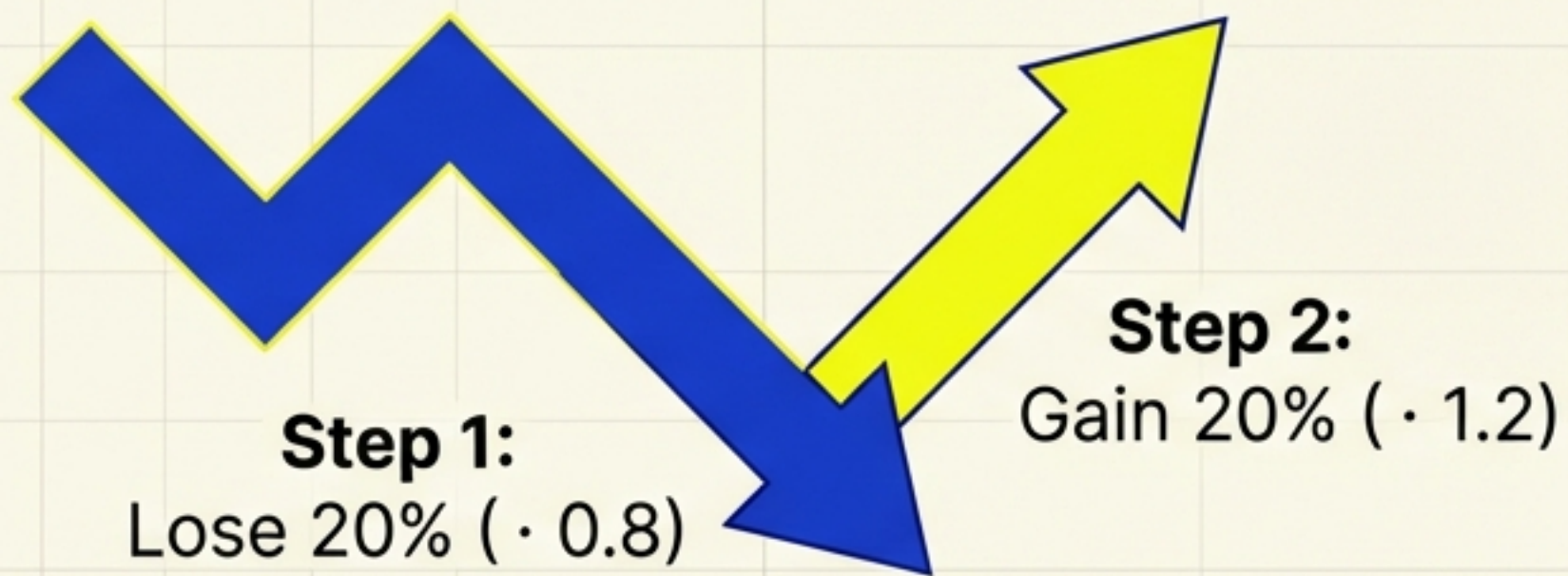
⚠ WARNING: Never subtract 20% from Brutto!

Real World: Growth & Fluctuations

Salary Duel



The Stock Trap



Start: 100%

Step 1: Lose 20% ($\cdot 0.8$)

Step 2: Gain 20% ($\cdot 1.2$)

Calculation: $0.8 \cdot 1.2 = 0.96$

Result: You are at 96%. You lost money.

Challenge Round: Rapid Fire

Increase BY 100%
vs. Increase TO
200%?



SAME THING.

(Factor 2)

Price drops by
50%.



It is HALVED.

(Factor 0.5)

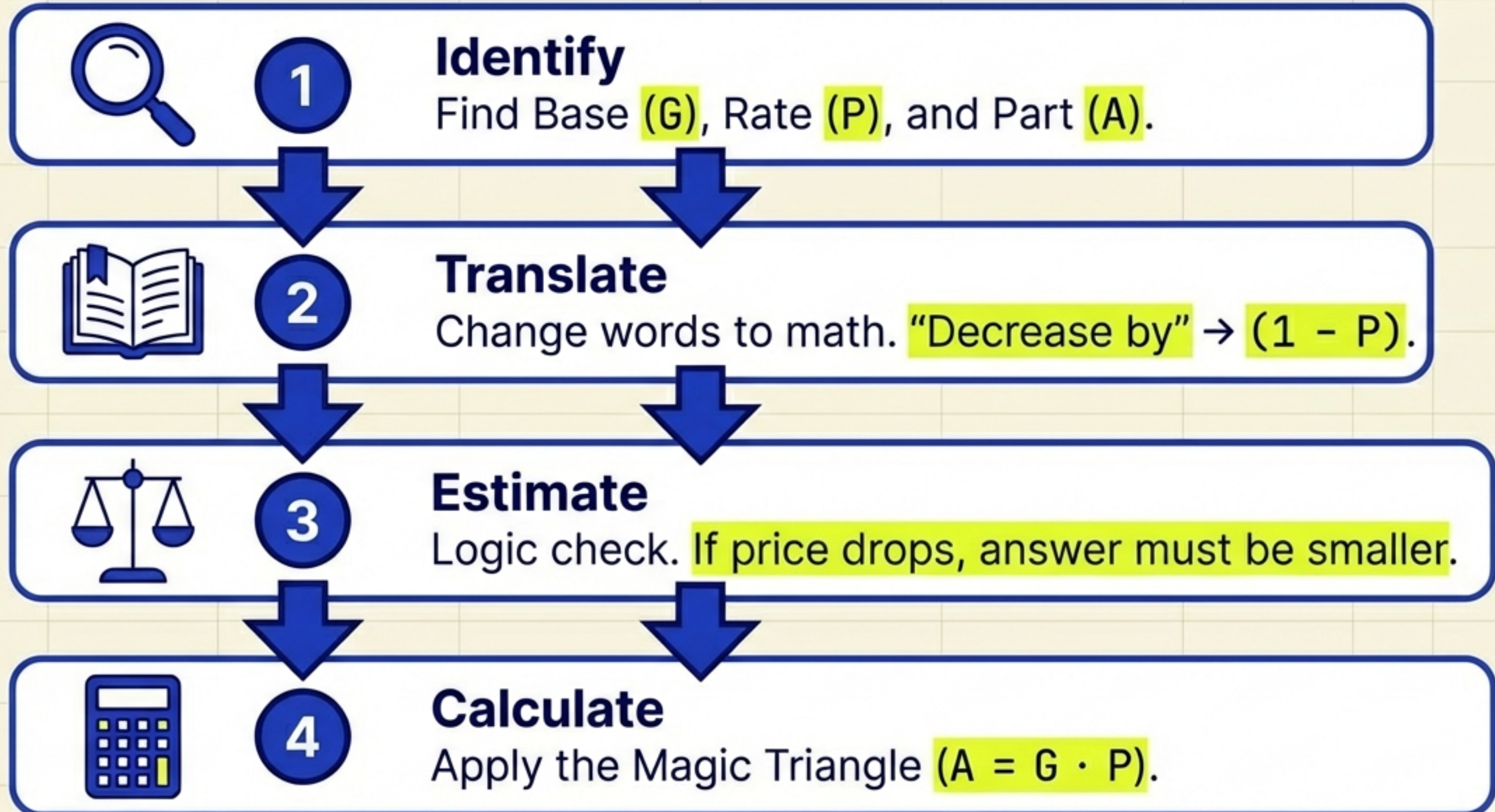
Price drops 50%,
then rises 50%.



75% of Original.

$(\$0.5 \cdot 1.5 = 0.75)$

Strategy: Cracking ANY Word Problem



The Cheat Sheet

$$A = G \cdot P \mid G = A/P \mid P = A/G$$

$$\begin{aligned} \text{Points} &= P_{\text{new}} - P_{\text{old}} \\ \text{Change} &= (P_{\text{new}} - P_{\text{old}}) / P_{\text{old}} \end{aligned}$$

Increase ($p\%$):
 $\cdot (1 + p/100)$

Decrease ($p\%$):
 $\cdot (1 - p/100)$

Good Luck on the LZK! You are ready. - VSN